

Visualizing the Impact of the Appalachian Mountains and the Surrounding Geography on Severe Weather in TN/NC

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Abstract— After Hurricane Helene made landfall in September 2024, it caused immense destruction not only to coastal states, but also to mountainous regions in East Tennessee and Western North Carolina. Despite being far from the coast, cities in these communities faced substantial flooding, damage, and deaths. The Appalachian Mountains and the geography in these areas have a large impact on weather fronts and wind velocities throughout this region. Using the Dyamond 3 dataset, we focused on the region from Knoxville, TN to Asheville, NC and created a visualization for observing wind velocities and weather patterns across this region during severe weather events.

Keywords— hurricanes, Appalachian Mountains, data engineering, dyamond dataset

I. INTRODUCTION

When Hurricane Helene made landfall over the Southern Appalachians in September of 2024, it brought forth one of the regions' most destructive storms in the form of catastrophic flooding, landslides, and hurricane-force wind gusts. This was especially true in the elevated regions near Asheville, NC. According to the National Hurricane Center (NHC), a stationary front near the Tennessee Valley was already a predecessor for rainfall in this area prior to Helene. The geography of the mountains in this region caused an increase of orographically driven upward vertical motion. These weather patterns resulted in maximized orographic effects as the hurricane made impact. Some regions received upwards of 30 inches of rain, devastating landslides, and around 107 fatalities in NC alone.[1][2].

This paper focuses on the impact of the Appalachian Mountains and their surrounding geography on severe weather in the NC and TN regions. Specifically, it investigates how wind, and especially vertical winds, vary from the Tennessee Valley up to the high elevation Smoky Mountains. We achieve this through visualizations of both the coast and the TN/NC border to plot various weather parameters from the Dyamond 3 dataset with specified elevations, locations, and time ranges. Our research questions are the following:

RQ1: How can the DYAMOND GEOS dataset be used to visualize wind and weather patterns in the TN/NC region?

RQ2: What impact does the geography of this region have on wind velocities? (Especially severe weather events).

II. METHODS

Our visualization is based on the DYAMOND 3 GEOS dataset through NASA [3]. The specific parameters that can be selected are wind velocity, pressure, and atmospheric temperature. To begin with, the Face 4 map is used in conjunction with the Python library Cartopy ([4]) for outlining the US states and specific cities (including Knoxville, TN and Asheville, NC). Timestamps from the DYAMOND dataset are converted to their Hour/Day/Month equivalent for more human readable times. This also allows the user to input a specific timeframe to capture the weather data.

Table 1: User Selectable Parameters

Parameters (Changeable)	Description
U	East-West Wind Velocity
V	North-South Wind Velocity
W	Vertical Velocity
Elevation [x,y]	Range of Elevation to view
Location (Coast or TN/NC)	Sets the view to either the entire Coast, or TN/NC region
Time selection (Day/Month)	Select Start/End dates

A. Program Features for Visualization of Data

After selecting the preferred parameters, the project features several methods of visualizing the results of the selected parameter

- Save a MP4 Video of the selected range
- Display the results with a set FPS within the notebook itself
- View a specific hour and day with set precision over the parameter values, elevation, and max/min color bar.
- Display the average velocities in Knoxville (the Valley) vs. Asheville (the Mountains) over the set range

III. RESULTS

To help answer our research questions, we focused on two regions: the South-East coast and the TN/NC border. Most of the time ranges tested were between September-October, during peak hurricane season in 2020 (also similar to the time range when Helene hit in 2024)

A. East-West Winds over the Coast

Using parameter “u”, we were able to pinpoint hurricanes in the Atlantic Ocean during September 17th, 2020. These correlate with storms Sally, Teddy, and Paulette.

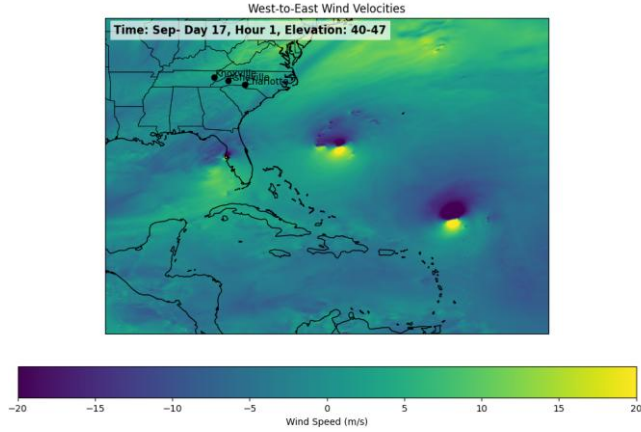


Figure 1: Hurricanes Visible on the Coast by Observing West/East Wind Velocities

As seen in Figure 1, this implementation is able to capture visible hurricanes in the Atlantic ocean by viewing the opposing wind directions (dark blue and yellow, respectively). This is done through location = “Coast” which gives a view of the outlined United States and shows the wind velocities in the nearby ocean.

B. Vertical Winds over the TN/NC Region

After viewing the incoming hurricanes during the September season, we changed the parameters to look at Vertical Winds (“w”) over the cropped TN/NC region.

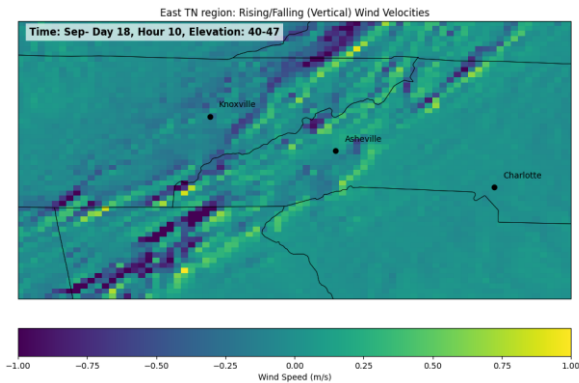


Figure 2: Vertical Wind Velocities through the Appalachian Region Post-Hurricane

As seen in Figure 2, the Appalachian mountains have a large impact on the vertical (upwards) during storm fronts. This image is taken on September 18th, around the time the hurricane would be impacting the NC/TN region. The yellow in the image represents fast rising air, which, according to

Marshall Shepherd, director of the Atmospheric Sciences Program at the University of Georgia, “forces [air] to rise over a mountain and cool, causing water vapor to condense”[5].

C. Mean Wind Velocities in Knoxville, TN vs. Asheville, NC

Lastly, we view the average wind velocities (specifically, Vertical velocities) over a range of time. This shows that the increase in wind over Asheville, NC isn’t just in a single hour’s screenshot, but that this region experiences higher wind gusts over regularly due to its geography over the Tennessee Valley.

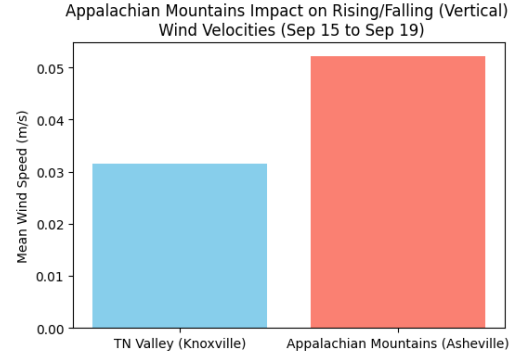


Figure 3: Mean Vertical Velocities between Knoxville and Asheville During/Post Hurricane

IV. CONCLUSION

The Southern Appalachian region has unique topography that have an impact on severe weather and wind velocities in this area. Through the DYAMOND dataset, we can visualize the increased wind gusts, especially orographic motion, to analyze the weather patterns that can lead to increased rainfall and severe weather in this region. According to Shepard, human-amplified climate change is increasing the temperatures in the Gulf, and therefore, increasing the strengths of storms (such as Hurricane Helene) [5]. This paper provides a visual tool for analyzing these regions and the impact of strong Hurricanes on the TN/NC region. This shows the importance of acknowledging the affects of climate change and being prepared to deal with severe weather in TN/NC during future hurricanes and storms.

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